

Friday, March 27, 2020

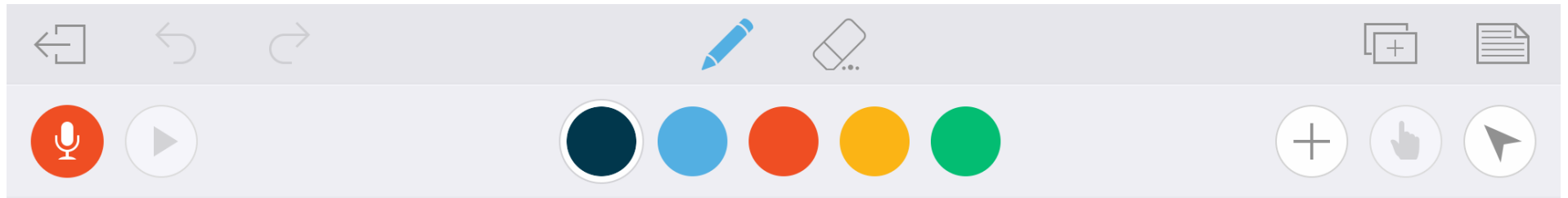


"Constant speed"

$$\vec{a} = 0$$

$$a_x = 0, a_y = 0$$

$$\sum F_x = 0, \sum F_y = 0$$

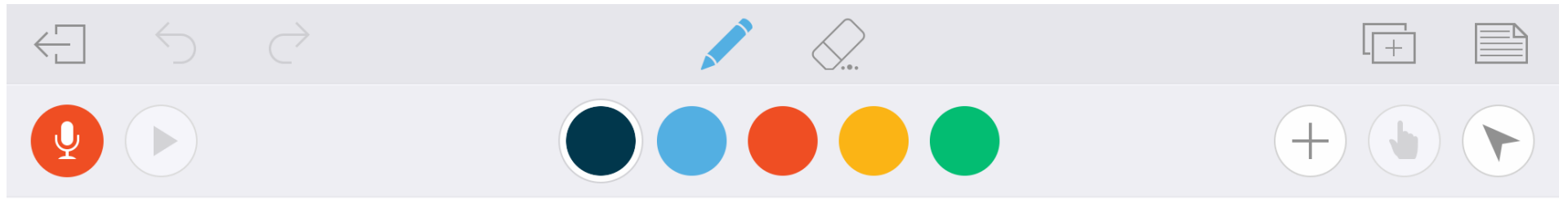


$$F - \check{m} \check{g} \sin \check{\theta} - f_k = 0$$

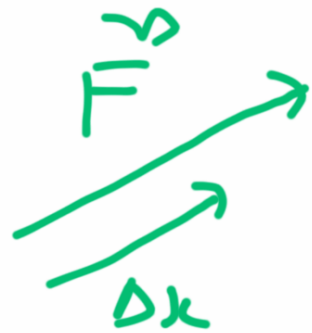
$$N - \check{m} \check{g} \cos \check{\theta} = 0$$

μ_k

$$f_k = \mu_k N$$



$$W_{\text{man/dote}} = \vec{F} \cdot \Delta \vec{r}$$

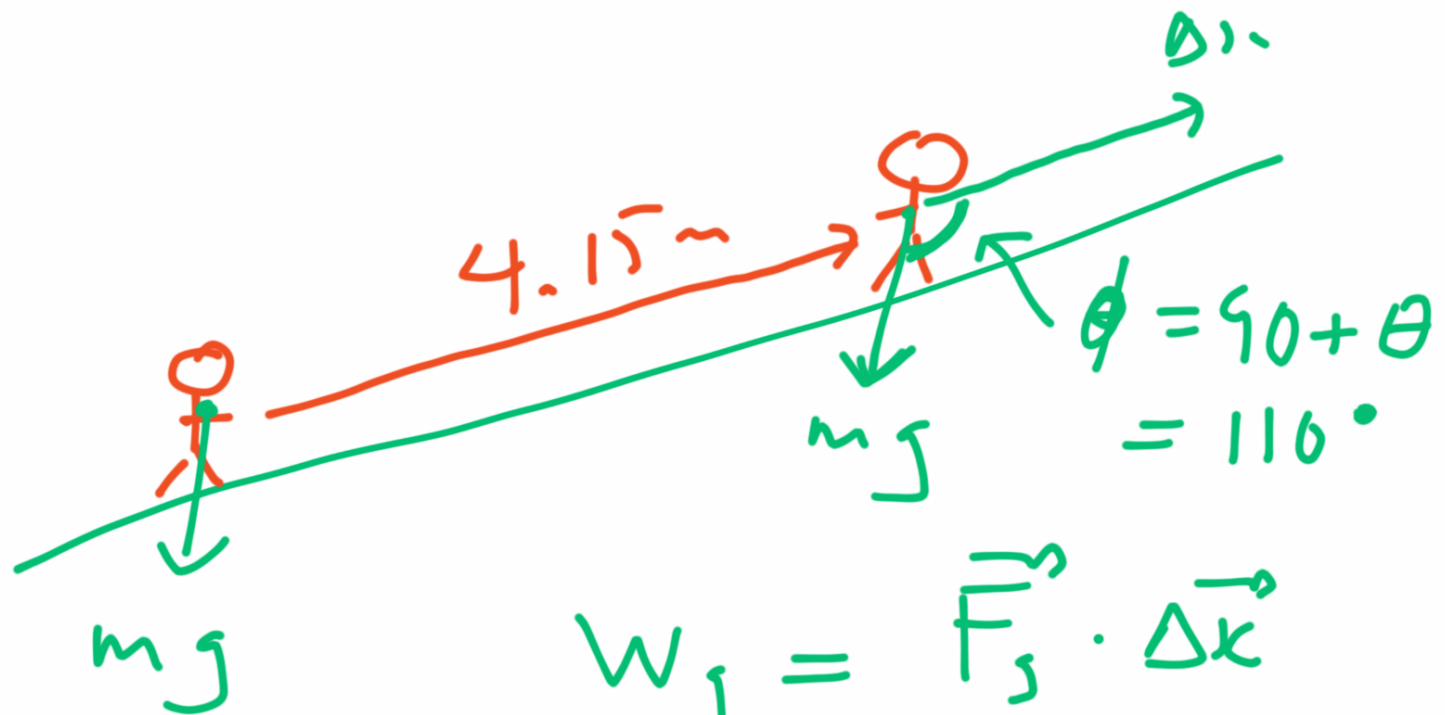
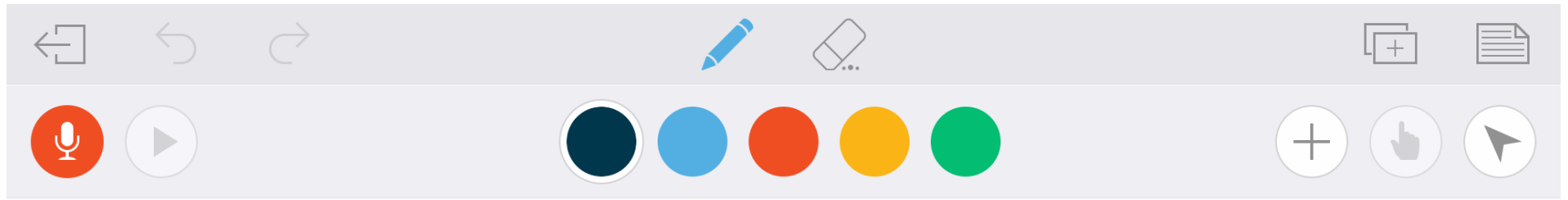


$$= |\vec{F}| |\Delta \vec{r}| \cos \theta$$

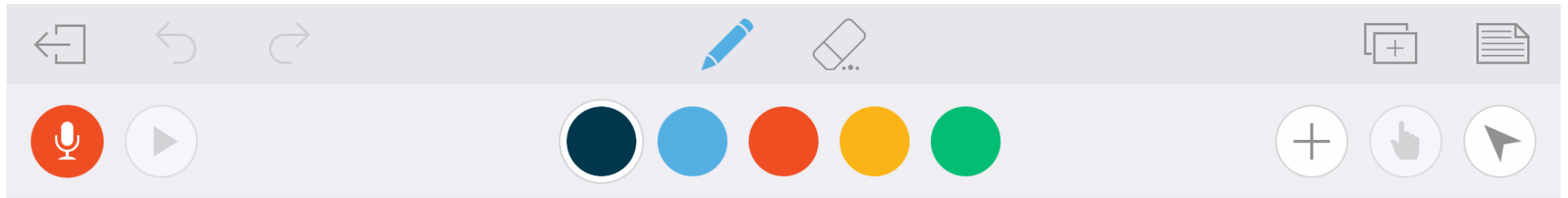
$$= |\vec{F}| |\Delta \vec{r}|$$

$$= (515)(4.15)$$

$$= 2137 \text{ J}$$



$$\begin{aligned} W_g &= \vec{F}_g \cdot \Delta \vec{r} \\ &= (m_g)(\Delta r) \cos \phi \\ &= (m_g)(\Delta r)(-0.342) \end{aligned}$$

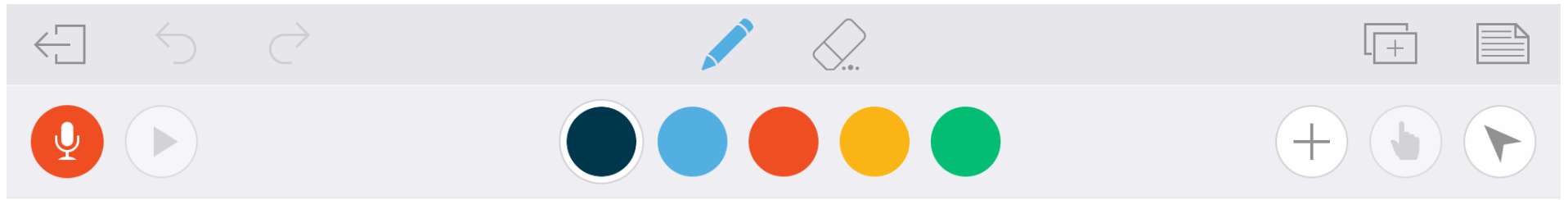


$$(80)(9.8)(4.15)(-.342) \\ = -1113 \text{ J}$$

$W_{\text{gas/overcome fric}}$

$$= 1113 \text{ J}$$

$$W_{\text{tot}} = \begin{array}{r} 1113 \text{ J} \\ + \\ 2137 \text{ J} \\ \hline = 3250 \text{ J} \end{array}$$

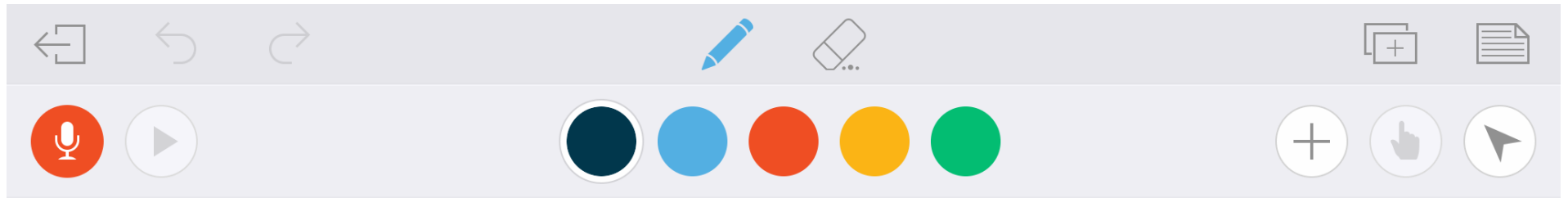


10.

$$K = \frac{1}{2} m v^2 = \frac{1}{2} (2004) (36.11)^2$$

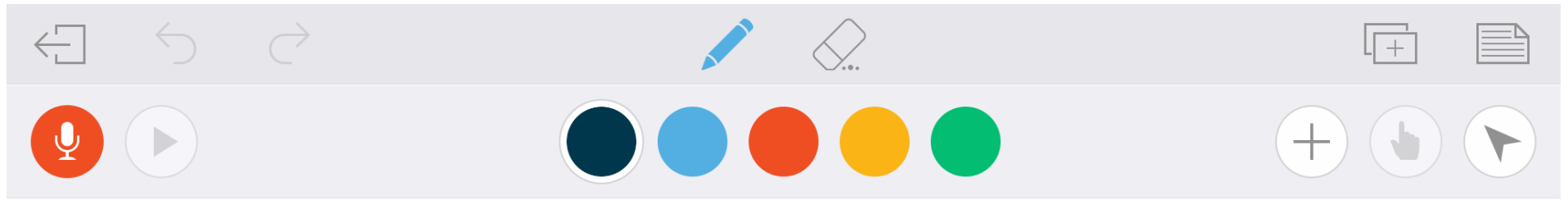
$$a) \quad v = 130 \frac{\text{km}}{\text{hr}} \times \frac{1000 \text{ m}}{\text{km}} = 1.31 \times 10^4 \frac{\text{m}}{\text{s}}$$

$$= 36.11 \frac{\text{m}}{\text{s}} \times \frac{1 \text{ hr}}{3600 \text{ s}}$$



$$\begin{aligned} v \cos \theta &= 27.4 \cos 40^\circ = v_2 \\ \underline{v \cos \theta} & \\ \text{constant} & \\ a_x &= 0 \\ a_y &= -g \\ K &= \frac{1}{2} m v_2^2 \\ &= \frac{1}{2} (-15) (v_2^2) \\ &= \underline{33.0 J} \end{aligned}$$

a)



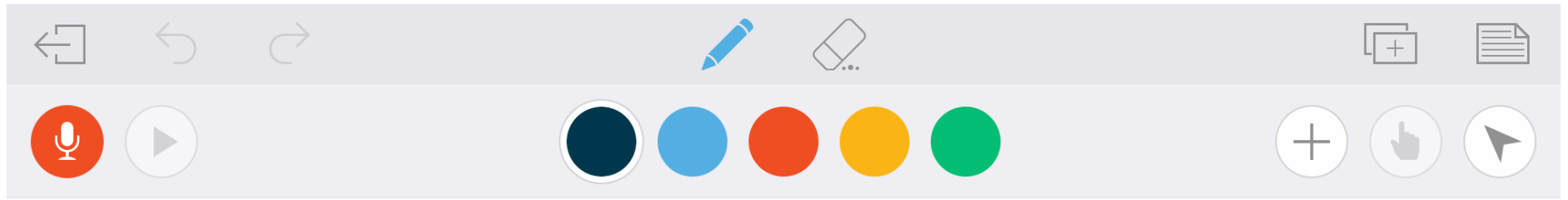
$$K_i = \frac{1}{2} m v^2 = \frac{1}{2} (.15)(\underline{\underline{27.4}})^2$$

(b) $\rightarrow$ $= 56.3 \text{ J} \checkmark$

$$K_f = 33.0 \text{ J} \checkmark$$

(c) $\Delta K = \underline{\underline{K_f}} - \underline{\underline{K_i}} = \underline{\underline{-23.3 \text{ J}}}$

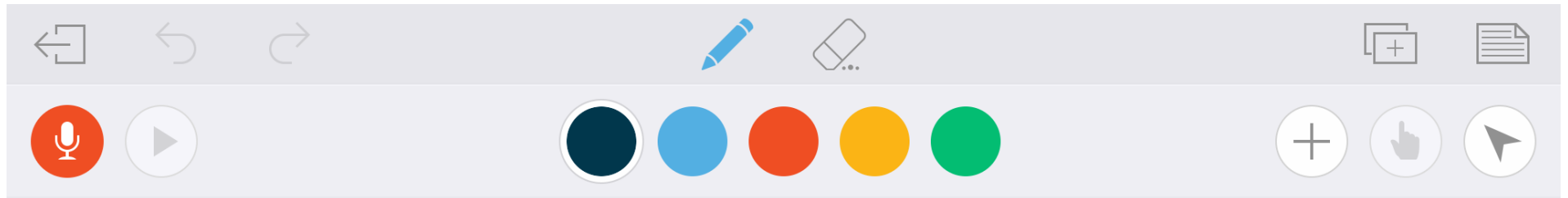
$$\underline{\underline{W_g}} = \underline{\underline{\Delta K}} = \underline{\underline{-mgh}}$$



d)

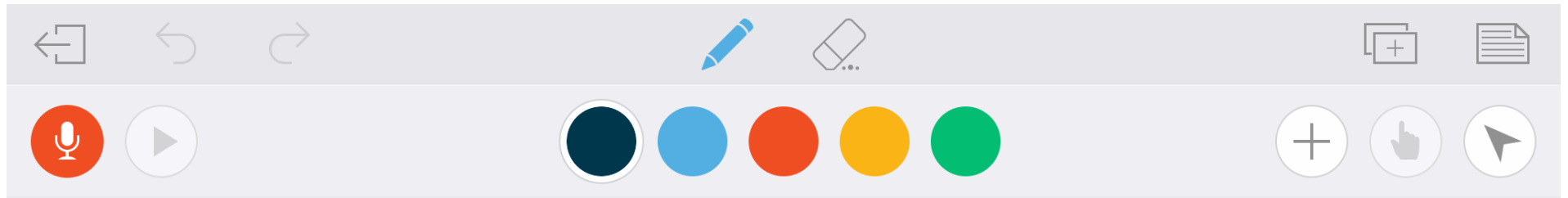
$$\begin{aligned} \frac{-mgh}{h} &= \frac{-23.3 \text{ J}}{-23.3 \text{ J}} \\ &= \frac{-(0.15)(9.8)}{15.9 \text{ m}} \end{aligned}$$

Final
Exam



$$h = 132 \sin 24^\circ$$

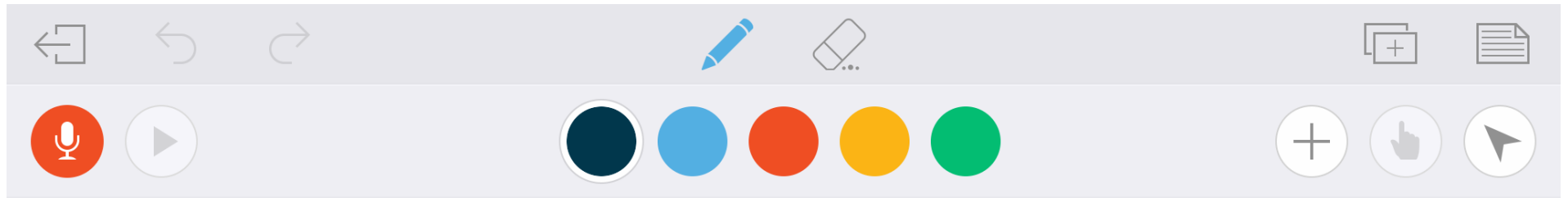
$$W_g = \underline{m} g \underline{h}$$



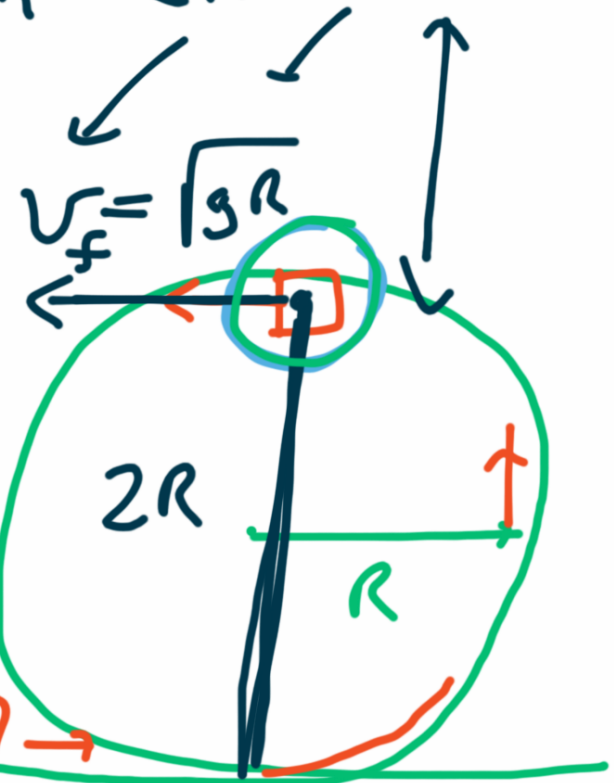
$$W_g = \Delta K$$
$$\cancel{m}gh = \frac{1}{2}\cancel{m}v_f^2 - \frac{1}{2}\cancel{m}v_i^2$$

$$v_f^2 = 2gh$$

$$v_f = \sqrt{2gh} = 32.4 \text{ m/s}$$



$$\Delta h = h - 2R$$



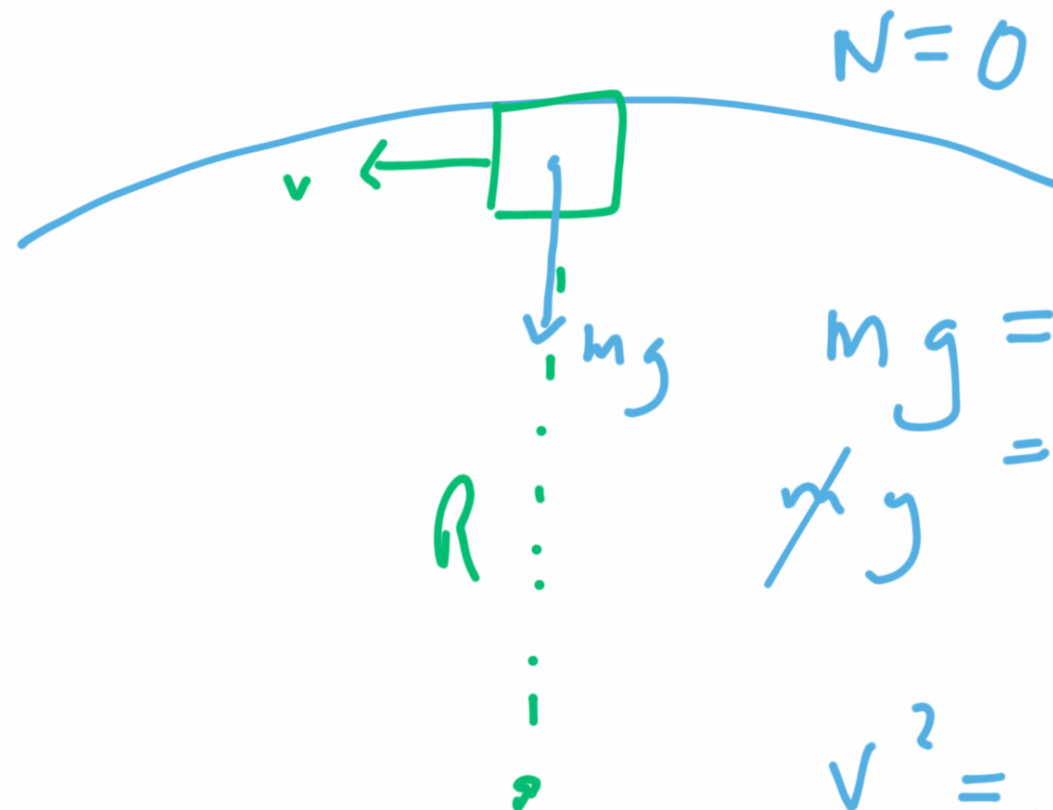
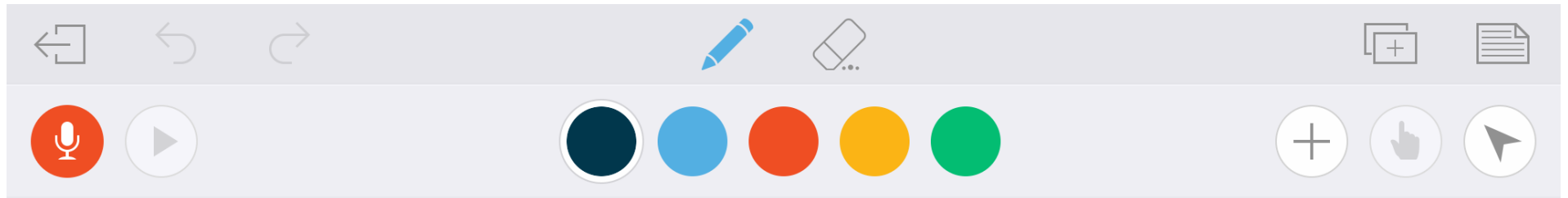
frictionless

Busch
Gardens.

$$\Delta K = K_f - K_i$$

$$= \frac{1}{2} m v_f^2 - \phi = \frac{1}{2} m (gR)$$

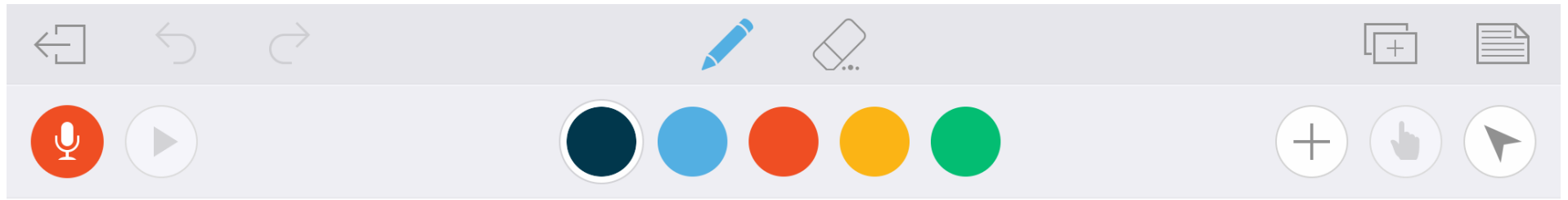
$$\Delta K = \frac{1}{2} m g R$$



$$mg = ma$$
$$\cancel{m}g = \cancel{m}\frac{v^2}{R}$$

$$v^2 = gR$$

$$v = \sqrt{gR}$$

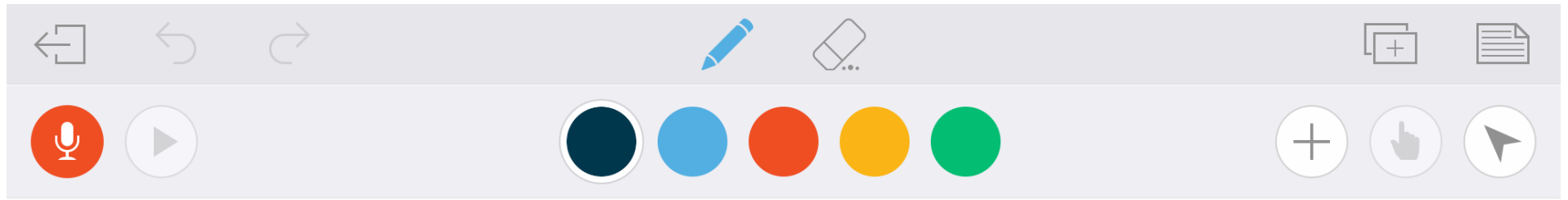


$$W_g = mg \text{ (change in height)}$$

$$= mg(h - 2R)$$

$$= \Delta K = \frac{1}{2} mg R$$

$$\cancel{mg}(h - 2R) = \frac{1}{2} \cancel{mg} R$$

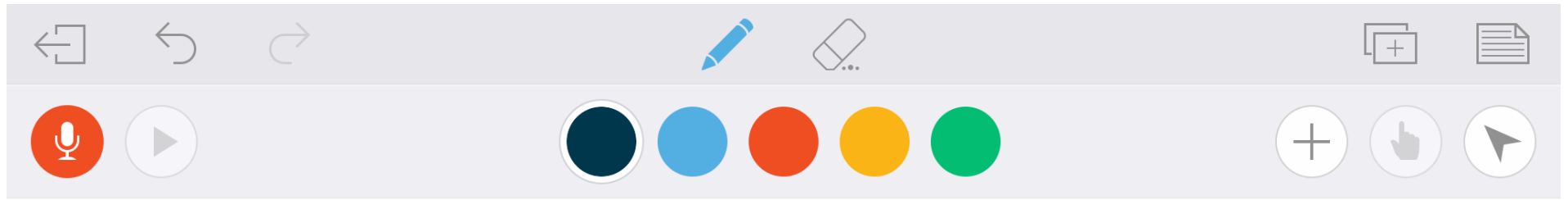


$$h - 2R = \frac{1}{2}R$$

$$h = 2R + \frac{1}{2}R$$

$$h = \frac{5}{2}R$$

✓✓



Due Date ; midnight
on Monday

Momentum & Collisions

+ Rotational Motion.